EXECUTIVE DECISION

Bring AI capacity into engineering

> **Acronyms and notation: AI**: artificial intelligence; **UI**: user interface.

A pilot of **six weeks** to reduce variable spending, retain sensitive context and maintain alternatives.

Local first.
Cloud when it adds value.

Scope: code, backend, UI and tests. Hybrid means local + cloud.

Planning proposal · 16 October 2026

Per-token consumption changes the bill

> **Acronyms and notation: US$**: United States dollars.

The team report points out users with spending of **$2,000–10,000**. We assume monthly amounts per professional here; verify against invoices.

US$ 10 thousand

5 people × US$ 2 thousand/month

US$ 50 thousand

5 people × US$ 10,000/month

This is not a market average. Cache, repeated context and parallel agents must appear in the baseline.

User-provided account · Arithmetic scenarios; no audited invoice

02 / OPPORTUNITY

Compact models deserve qualification

Acronyms and notation: SWE: software engineering; benchmark family for software tasks; **B / A…B**: B denotes billions of parameters; A before the count denotes active parameters per token.

Benchmark	Qwen3.6 · 27B	Qwen3.8 · 27B
Terminal Bench 2.1	63.4	**73.0**
SWE-bench Pro	53.5	**61.7**
NL2Repo	36.2	**42.3**

The same nominal parameter count, with a higher published result. Validate quantization and internal tasks; the dataset was not isolated as the sole cause.

Qwen · evaluation of the authors themselves

03 / BUSINESS EVIDENCE

Hybrid savings depend on the task

Task	Cloud → hybrid	Change
Ktor mapper	$0.339 → $0.134	−60.5%
TypeScript rename	$0.227 → $0.121	−46.7%
Spring field	$0.092 → $0.164	+78.3%

Local Qwen on Mac + remote Sonnet. Select task families using accepted-task cost; these are not monthly team savings.

Start with 32, 48 and 64 GB too

Acronyms and notation: GB: gigabyte, a decimal unit of data capacity; **MoE**: mixture of experts; activates a subset of expert networks per token; **MLX**: Apple machine-learning framework; runtime name, not a quantization; **oMLX**: model-serving software built on MLX; **MTP**: multi-token prediction; predicts multiple tokens for speculative generation; **Q4**: 4-bit quantization family; suffixes identify the specific format; **Q8**: 8-bit quantization family; suffixes identify the specific format; **tok/s**: tokens per second; a throughput measure; **B / A…B**: B denotes billions of parameters; A before the count denotes active parameters per token.

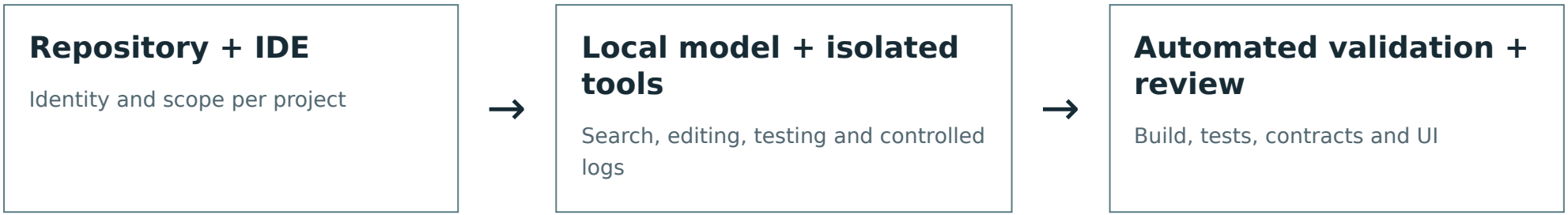
Mac	Model / runtime	Decode tok/s
M1 Pro · 32 GB	35B MoE Q4 · llama.cpp	25.4
M4 Pro · 48 GB	27B Q4 · Rapid-MLX	15.2
M4 Pro · 64 GB	27B Q4 / Q8 · oMLX + MTP	20.8 / 16.6

Reuse workstations before purchasing. Different protocols; the 32 GB test records host swap. Memory alone does not determine speed or quality.

M1 Pro 32 GB · M4 Pro 48 GB · M4 Pro 64 GB

Preserve context. Escalate according to policy.

Acronyms and notation: UI: user interface; **IDE**: integrated development environment.



Authorised exceptions: the gateway minimizes context and calls the cloud. Data prohibited from leaving stays local.

Proposed architecture · fallback never replaces authorization to send data

06 / SECURITY

Internet access with enforced boundaries

Isolated agent worker → authenticated local model → disposable tests.

Documentation through a restricted broker. Cloud export through a separate authorization gate.

No production credentials, broad host mounts or unrestricted shell egress. Review generated workflows and hooks as well as code.

Docker security model · Sandbox runtime

Qualify OMP, OpenCode and Aider

Acronyms and notation: MLX: Apple machine-learning framework; runtime name, not a quantization; **OMP**: Oh My Pi; coding agent; **VM**: virtual machine; **OS**: operating system.

Layer	Decision
MLX / llama.cpp	Private, authenticated inference gateway
Agent	Restricted effective tool policy
Execution	OS/VM boundary and controlled network
Acceptance	Independent tests and human review

Open-weight deployment is manageable. The agent name alone is not a security guarantee.

OpenCode · OMP · Aider

07 / SECURITY VALIDATION

Prove containment before expansion

Test malicious repository instructions, web redirects, synthetic-secret export, dependency scripts and access outside the worktree.

Verify that project configuration cannot weaken corporate policy and that cloud fallback cannot bypass data restrictions.

Evidence: denied operations, network traces, accepted patches and bounded execution. Physical side channels remain a specialist appendix.

Isolation layers and documented limitations

Use the cloud to improve reusable local workflows

Cloud optimizer proposes a task-specific harness. Independent tests select a version. The local model executes recurring work.

Version instructions, retrieval, tools and bounded recovery; preserve rollback and untouched acceptance tests.

SkillOpt supports optimizing skills without changing weights. Coding savings must still be measured on the actual repository.

Microsoft SkillOpt

Start with verifiable tasks

> **Acronyms and notation: QA**: quality assurance; **UI**: user interface; **API**: application programming interface; connects software components; **E2E**: end-to-end tests covering the complete workflow; **SQL**: Structured Query Language; database query language.

Workstream	Initial application	Gate
Backend	APIs, SQL and Delimited Refactoring	Contracts + tests
QA	Cases, fixtures and regressions	Execution + useful coverage
UI	Components and visual corrections	Captures + accessibility + E2E
Legacy	Explanation and changes per module	Behavior characterization

Accepting code requires evidence of execution, even when the agent says it has finished.

Proposed engineering plan; internal rates will be measured during the pilot

10 / INVESTMENT

US$ 41,400 to qualify the approach

Acronyms and notation: US$: United States dollars.

Component	Assumption	US$
Workstations	5 × 5,000	25,000
Integration	80 h × 100	8,000
Security + validation	30 h × 100	3,000
Contingency	15%	5,400

Estimated additional operation: **US$ 500/month**. Configurations and prices still require quotations.

Planning scenario, not a commercial quotation from a supplier

The avoided portion determines payback

Acronyms and notation: API: application programming interface; connects software components; **US$**: United States dollars.

API/person/month	Team savings / month	Payback
US$ 2,000	US$ 4,500	9.20 months
US$ 5,000	US$ 12,000	3.45 months
US$ 10,000	US$ 24,500	1.69 month

5 users · 50% of invoice avoided · $500/month of operation · full investment of US$ 41,400.

Sensitivity: in the first scenario, avoiding only 20% leads to a return to 27.6 months.

Own calculation · additional rework must be discounted · calculator in the proposal

12 / PILOT

Six weeks, with criteria defined in advance

Acronyms and notation: pp: percentage points when comparing percentages; lowercase pp in benchmark commands denotes prompt processing.

1

Baseline

Invoices, tasks, quality and review

2

Implementation

Configuration, isolation and controls

3–5

Comparison

Real tasks, current flow versus hybrid

6

Decision

Savings, quality, security and capacity

Proposed targets: ≥30% total savings; ≤5 pp loss of acceptance; ≤10% increase in cycle; no exposure in defined tests.

Criteria for negotiation; these are not results already achieved

13 / RECOMMENDATION

Buy evidence before scaling

> **Acronyms and notation: QA**: quality assurance; **FinOps**: financial operations practice for managing technology consumption and cost.

Approve the pilot, confirm the baseline and make the expansion result dependent.

Cost per accepted task.
Data governed by policy.
Versions under control.

Platform + Security + FinOps + Engineering/QA share the decision.

Next proposed step · no purchase, hiring or automatic publication

From decision to detail

Acronyms and notation: SSD: solid-state drive; **GB**: gigabyte, a decimal unit of data capacity; **MLX**: Apple machine-learning framework; runtime name, not a quantization; **PDF**: Portable Document Format.

Full research

Business cases, Macs tables 16–512 GB, MLX/llama.cpp, SSD, experts and security.

Open Report →

Investment proposal

Problem/Context, Solution, Required Investment and sensitivity calculator.

Open Proposal →

Sources appear in the footnotes and documents. Arrow keys navigate between slides; “Print / PDF” exports the entire presentation.

Offline package · 16 October 2026

TECHNICAL SUPPORT / SSD

Larger models on the same workstations

Acronyms and notation: CPU: central processing unit; **SSD**: solid-state drive; **GB**: gigabyte, a decimal unit of data capacity; **IQ2_K**: low-bit quantization format used for model weights; **WARP**: Weight-Aware Runtime and Paging; weight-streaming inference runtime; **tok/s**: tokens per second; a throughput measure.

Mac / model	SSD path	Decode tok/s
32 GB M1 Max · Flash-Next	MoEspresso · mainly IQ2_K	12–15
48 GB M5 Pro · Flash-Next	Slotstream · 4-bit	~12
64 GB M5 Pro · Flash-Next	llama.cpp fork / Slipstream	27.6 / 35.0–37.5
64 GB M5 Pro · DeepSeek V4.1	WARP · CPU	3.77 / 3.71

Authors’ results, without a common protocol. MoEspresso changes routing. WARP measures 64 / 200 tokens; it does not establish the 20–30 target.

MoEspresso · Slotstream · 64 GB · WARP · Slipstream

DECISION SUPPORT / INVESTMENT

Smaller purchase, the same economic discipline

> **Acronyms and notation: GPU**: graphics processing unit; **RAM**: random-access memory; working memory; **VRAM**: video memory available to the graphics processor; **SSD**: solid-state drive; **GB**: gigabyte, a decimal unit of data capacity; **US$**: United States dollars.

Compare reuse with **US$ 2500–5000** workstations: five units total US$ 12500–25000 in hardware. Planning amounts; obtain configuration and SSD quotations.

The home account of ~60 tokens/s uses a 32 GB VRAM GPU + 64 GB RAM + SSD; logs and full configuration are still unavailable. It is not a Mac benchmark.

Measure four paths: cloud, generic local, local with harness, and local with harness + cloud. Approve based on net savings, quality, time and data containment.

Pilot proposal · account without independent reproduction · TCO includes integration and operation